Quantum Mechanics I — Exercise Sheet 4

Many levels

Variant: Questions only

Conventions, as in Lecture 4: the three-level system has the orthonormal basis e_+, e_0, e_- and the Hamiltonian $H = \varepsilon(|e_+\rangle\langle e_+| + |e_-\rangle\langle e_-|)$, $\varepsilon > 0$, with the projections $E_0 = |e_0\rangle\langle e_0|$ and $E_\varepsilon = |e_+\rangle\langle e_+| + |e_-\rangle\langle e_-|$; $Z = \operatorname{diag}(1, 0, -1)$, $X = |e_+\rangle\langle e_-| + |e_-\rangle\langle e_+|$ and $\chi_\pm = \frac{1}{\sqrt{2}}(e_+ \pm e_-)$.

Exercise 1 (a state of the pair, read coarsely and finely). Let $\psi = \alpha e_+ + \beta e_0 + \gamma e_-$ be a unit vector with $|\alpha|^2 + |\gamma|^2 > 0$.

1. The energy is measured and reads ε. Show that a reading of X made at once then gives $+1$ with probability $|\alpha + \gamma|^2 / \big(2(|\alpha|^2 + |\gamma|^2)\big)$.
2. Instead, Z is measured and only "not 0" is recorded. Show that X then gives $+1$ with probability $\frac{1}{2}$, and that the two answers differ exactly when $\operatorname{Re}(\bar\alpha\gamma) \neq 0$.
3. Show that if the reading of X is made at a later time t instead, with the system evolving under H in between, nothing changes.
4. Take $\psi = \frac{1}{3}(2e_+ + e_0 + 2ie_-)$. Find the state after the reading ε, and an observable that tells the two experiments of (i) and (ii) apart although X does not.

Exercise 2 (three equally spaced levels). On $\mathbb{C}^3$ with the orthonormal basis e_1, e_2, e_3, let $H = \operatorname{diag}(0, \varepsilon, 2\varepsilon)$, $\varepsilon > 0$, and $\psi = \frac{1}{\sqrt{3}}(e_1 + e_2 + e_3)$. Put $u = \varepsilon t/\hbar$.

1. List the positive Bohr frequencies of H.
2. Show that the probability that a measurement of $|\psi\rangle\langle\psi|$ at the time t reads 1 is $\frac{1}{9}(3 + 4\cos u + 2\cos 2u)$. Find its smallest value and the first time it is reached.
3. For $A = |e_1\rangle\langle e_3| + |e_3\rangle\langle e_1|$, show that the mean of A in the state ψ_t is $\frac{2}{3}\cos 2u$. Which Bohr frequency is missing, and why?

Exercise 3 (stationary density matrices).

1. Let $H = \sum_k \varepsilon_k E_k$ be a constant Hamiltonian on $\mathbb{C}^n$. Show that every stationary density matrix is the density matrix of an ensemble of energy eigenvectors.
2. For the Hamiltonian of Lecture 4, describe all the stationary density matrices.
3. Show that $\rho = \frac{1}{3}\big(|e_0\rangle\langle e_0| + |\chi_+\rangle\langle\chi_+| + |e_+\rangle\langle e_+|\big)$ is stationary, although it is not diagonal in the basis e_+, e_0, e_-, and write it as an ensemble of energy eigenvectors.

Exercise 4 (the unread energy reading). The energy of an n-level system is measured and the reading is not recorded, so that, by (3.8), $\rho \mapsto \rho' = \sum_k E_k \rho E_k$.

1. Show that $\operatorname{Tr}\rho^2 - \operatorname{Tr}\rho'^2 = \sum_{k\neq l} \operatorname{Tr}\big(E_l \rho E_k \rho\big) \geq 0$.
2. Find ρ' and $\operatorname{Tr}\rho'^2$ for the state $\psi = \frac{1}{\sqrt{3}}(e_+ + e_0 + e_-)$ of Example 4.8.
3. Show that ρ' is always stationary, and that a second unread energy reading leaves it unchanged.

Exercise 5 (the same means, with the observables moving). Take the system of Exercise 2: $H = \operatorname{diag}(0, \varepsilon, 2\varepsilon)$, $\psi = \frac{1}{\sqrt{3}}(e_1 + e_2 + e_3)$, $A = |e_1\rangle\langle e_3| + |e_3\rangle\langle e_1|$ and $u = \varepsilon t/\hbar$. Let $A_t = U_t^\dagger A U_t$ (Proposition 4.5).

1. Find A_t.
2. Check that $\mathrm{d}A_t/\mathrm{d}t = \frac{\mathrm{i}}{\hbar}[H, A_t]$.
3. Compute $\langle \psi, A_t \psi \rangle$ and compare it with Exercise 2 (iii).
4. Show that the projection $P = |\psi\rangle\langle\psi|$ moves to $P_t = \left|U_t^\dagger \psi\right\rangle\!\left\langle U_t^\dagger \psi\right|$, and that its mean in the state ψ is the probability of Exercise 2 (ii).
5. Of A, P and $|e_2\rangle\langle e_2|$, which stay fixed for all t, and why?

Exercise 6 (⋆ conditioning changes the weights, for every weight). Take the ensemble $\{(p, e_+), (1-p, \frac{1}{\sqrt{2}}(e_0 + e_-))\}$ with $0 < p < 1$, and measure the energy. Find the probability of the reading ε. Given that reading, find the weights of the members afterwards, and the probability that Z then reads +1. Show that keeping the old weights would give a different answer for every p.